

Expert Thermal / XThermal

Task-1 -Thermal Model Analysis

1.Assumptions and their justifications

The physics-based thermal model for the heat sink design incorporates several assumptions:

- **Air Properties:** Assumes constant air properties (thermal conductivity $k_{\text{air}} = 0.0262$ W/m.K, kinematic viscosity $\nu_{\text{air}} = 1.57\text{e-}5$ m²/s, Prandtl number $Pr_{\text{air}} = 0.71$) at 25°C. This simplifies calculations but may not hold true for varying ambient conditions.
- **Flow Regime:** The Nusselt number calculation distinguishes between laminar (Reynolds number < 2300) and turbulent flow. This is a standard simplification for heat transfer in channels.
- **Idealized Heat Sink Geometry:** Assumes a uniform fin thickness and spacing. Complex geometries or manufacturing imperfections are not considered.
- **Uniform Heat Flux:** The model implicitly assumes uniform heat generation across the die area.
- **Constant Thermal Interface Material (TIM) Properties:** k_{tim} and t_{tim} are treated as constant, whereas in reality, TIM properties can vary with temperature and pressure

2.Key results from the model

- **Physics Model Validation:** For an example scenario (TDP=180W, Air Velocity=7m/s, $k_{\text{TIM}}=8.0$ W/m.K), the model predicts a total thermal resistance (R_{total}) of approximately **0.288 °C/W** and a junction temperature (T_{j}) of approximately **76.88 °C**.
- **Dataset Generation:** A comprehensive dataset was generated by sweeping TDP (30-250W), Air Velocity (0.5-15 m/s), and k_{TIM} (1-12 W/m.K), resulting in **3960 unique data points** for training ML surrogates.

3. ML model accuracy comparison

Two ML models, RandomForest Regressor and a Neural Network (MLP Regressor), were trained to act as surrogates for the physics model. Their performance on the test set is as follows:

Model	MAE	RMSE	R ²
RandomForest	0.09018	0.28400	0.994962
NeuralNet_MLP	0.35848	0.62736	-1.780435

- The **RandomForest Regressor** significantly outperformed the Neural Network, demonstrating excellent accuracy with a very high R² value close to 1, indicating it captures almost all the variance in the target variables (R_{th} and T_{junction}).
- The **NeuralNet_MLP** performed poorly, with a negative R² value, suggesting it is a worse predictor than simply using the mean of the target values.

4. Important Correlations or Sensitivities

- **Correlation with T_{junction}:**
 - TDP_W: Very strong positive correlation (0.963), indicating that increasing power significantly increases junction temperature.
 - AirVel_{ms}: Moderate negative correlation (-0.192), implying higher air velocity leads to lower junction temperatures.
 - k_{TIM}: Weak negative correlation (-0.056), suggesting better TIM conductivity has a minor impact on junction temperature in this model.
 - R_{th}: Weak positive correlation (0.243).
- **RandomForest Feature Importance:**
 - TDP_W: 0.926
 - AirVel_{ms}: 0.068
 - k_{TIM}: 0.006

These values reinforce that **TDP (Thermal Design Power)** is by far the most critical factor influencing the junction temperature, followed by air velocity. The thermal conductivity of the TIM (k_{TIM}) has a relatively minor impact within the simulated range.

5. Limitations of the approach

- **Simplified Physics Model:** The heat sink model is a simplified representation. It does not account for complex phenomena like non-uniform airflow, radiation heat transfer, temperature-dependent material properties, or detailed fin efficiency calculations beyond basic convection.
- **Data Range:** The dataset covers specific ranges for TDP, Air Velocity, and k_{TIM} . Extrapolating the ML models outside these ranges may lead to inaccurate predictions.
- **Neural Network Performance:** The current MLP model performed very poorly, suggesting it either requires significant hyperparameter tuning, a different architecture, or the data distribution is not well-suited for its current configuration.
- **Single Heat Sink Design:** The analysis is based on one specific heat sink geometry. The findings might not be directly transferable to other designs.

6. Areas for Improvement or Optimization

- **Advanced Physics Model:** Incorporate more detailed heat transfer mechanisms (e.g., radiation, non-uniform heat generation, more sophisticated fin efficiency models) to enhance the accuracy of the physics model.
- **ML Model Tuning:** For the Neural Network, conduct extensive hyperparameter tuning (e.g., number of layers, neurons per layer, activation functions, learning rate, regularization) to improve its performance. Consider other advanced ensemble methods or deep learning architectures.
- **Expanded Dataset:** Generate a more diverse dataset by varying additional heat sink design parameters (e.g., fin height, base thickness, fin material) to make the ML surrogates more generalizable.
- **Uncertainty Quantification:** Implement methods to quantify the uncertainty in the predictions of the ML surrogates, providing a range of possible outcomes rather than just point estimates.
- **Optimization Framework:** Integrate the accurate RandomForest surrogate model into an optimization loop to rapidly explore design parameters and identify optimal heat sink configurations for desired thermal performance targets (e.g., minimum R_{th} for a given $T_{junction}$ and manufacturing constraints).

7.You may also include GitHub, portfolio, or demo links if available.

Link-https://github.com/Yasmeen-Begum/ET_Assessment_AI_26/tree/main/Deliverables

Task 2: Agentic AI / Workflow AI Understanding In your own words, (No AI assistance)

Please explain your understanding of:

- **Agentic AI**

Agentic AI is a type of AI system that can act autonomously make decision, plan an action, execute tasks, produce results, provide reasons with minimal human intervention. It follow think, act, observe pattern.

- **Workflow AI**

AI workflows automate related tasks across industries from sales and marketing to software development ,by integrating AI tools to generate analyze and process data efficiently.

Example: n8n

- **AI copilots / assistants**

They are the chatbot type with AI model integrated backend from google-gemini models, Aws-Titan models ,Anthropic-Claude models where they can give responses based on the prompt enter by the user .It provides detailed instructions related to text, code, images and generate data to the user.

- **Difference between a standalone AI model and AI integrated into a product workflow**

-Standalone AI Model: A specialized application used to perform particular AI-enabled activities, such as ChatGPT or Claude.

-AI Embedded in a Product Workflow: AI functionalities embedded within an already established software or device, such as Grammarly on Google Docs or Microsoft Copilot on Office applications.

Also describe any past experience you have with:

-GenAI

Link-

<https://github.com/YasmeenBegum/Hack2Skill/tree/main/Gen%20AI%20Academy%20APAC%20Edition/Multi-Agent%20Productivity%20Assistant>

-LLMs,

[https://github.com/Yasmeen-](https://github.com/Yasmeen-Begum/Udacity/tree/main/2.Introducing%20Generative%20AI%20with%20AWS/Project%3A%20Building%20a%20Domain%20Expert%20Model)

[Begum/Udacity/tree/main/2.Introducing%20Generative%20AI%20with%20AWS/Project%3A%20Building%20a%20Domain%20Expert%20Model](https://github.com/Yasmeen-Begum/Udacity/tree/main/2.Introducing%20Generative%20AI%20with%20AWS/Project%3A%20Building%20a%20Domain%20Expert%20Model)

-RAG

<https://github.com/Yasmeen-Begum/RagAssitant>

-AI Agents

<https://github.com/Yasmeen-Begum/Pharmaceuticals>

- Workflow automation -n8n I have practiced
- AI integration into a web platform or product If you do not have direct experience, explain how you would approach learning and implementing these concepts.

I have learned if you can provide documentations links, videos links , github pages,notebooks

Task 3: AI Use Case Evaluation

Suggest 3–5 AI features that could be useful in an engineering software platform. For each feature, briefly mention:

- What problem it solves
- What data would be required;
- Whether it requires ML, GenAI, workflow AI, or a combination;
- How you would validate that it works correctly. Examples of possible areas include:
- AI-assisted input validation
- Missing parameter detection
- Result summarization
- Thermal design recommendations
- Report generation
- Knowledge retrieval from engineering documents
- ML surrogate models for fast prediction
- Physics-aware AI for future thermal analysis

Feature/Aspect	AI-Assisted Input Validation & Missing Parameter Detection	ML Surrogate Models for Fast Prediction	Thermal Design Recommendations	Intelligent Report Generation & Result Summarization
Primary AI Type(s)	Workflow AI, ML	ML	GenAI, ML, Workflow AI	GenAI, ML, Workflow AI
Problem Solved	Prevents input errors and identifies missing data	Accelerates design iterations by replacing expensive simulations	Guides engineers toward optimal thermal solutions	Automates reporting and summarizes complex simulation results
Key Data Required	Historical inputs, engineering rules, templates	High-fidelity simulation datasets	Design database, domain knowledge, simulation results	Structured outputs, design parameters, historical reports
Validation Method	Unit tests, A/B testing, expert review	MAE, RMSE, R ² , cross-validation, physics comparison	Expert review, simulation validation, user feedback	Manual report comparison, user acceptability, fact-checking

Task 4: Motivation and Ownership

Please answer the following(In your own words):

- Why do you want to join Expert Thermal / XThermal?

My desire to be a part of Expert Thermal / XThermal stems from the fact that it allows the use of AI in the implementation of real-world engineering problems rather than creating generic AI applications

- What are you most interested in building in AI/ML?

I have started my career in the domain of AIML because it help me to learn real world problem understanding and model building related to daily life scenarios .Now it is a fastest growing field.

- Are you more interested in ML models, GenAI workflows, physics-aware AI, or product AI integration?

I have interested building in all domains

- How do you handle ambiguous problems where requirements are not fully defined?

I will see task related codes in google or AI copilots, github pages

- Are you comfortable working in a fast-moving startup environment?

Yes

- How do you think you can grow within our team?

I can learn and adapt , try to make me involve and understand your company team in working environment.

- This role calls for ownership, commitment, and initiative. How would you bring new ideas, create solutions, and take on greater responsibility over time?

In my opinion, ownership really is about being accountable for the end result of a solution, not just doing one's little piece of the work. I am a great fan of discovering new tools, finding peerless ways of doing things, and perfecting existing workflows time and again.